

Study Plan

Perfect – here's **(1) a week-by-week study plan** and **(3) a direct mapping to India job descriptions**, optimized for **Market Risk first, MRM second**, assuming **no finance background but strong stats/programming**.

Week-by-Week Study Plan (16 Weeks)

This is **realistic, industry-aligned**, and avoids CFA-style fluff.

Phase 0 - Financial Intuition (Weeks 1-2)

Week 1: How Markets Create P&L

Concepts

- Prices → returns → P&L
- Long vs short
- Linear instruments (stocks, bonds, FX)

Hands-on

- Download any price series (equity index or FX)
- Compute daily returns
- Plot price vs returns

Deliverable

- Notebook: *"Price vs Returns intuition"*
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Week 2: Portfolios & Risk Intuition

Concepts

- Portfolio weights
- Diversification
- Why correlation matters

Hands-on

- 2-3 asset portfolio
- Portfolio returns

- Correlation matrix

Deliverable

- Notebook: *"Portfolio return & correlation"*

📌 **At this point, finance stops being scary.**

Phase 1 - Market Risk Modeling (Core) ⭐⭐⭐ (Weeks 3-10)

Week 3: Return Distributions

Concepts

- Simple vs log returns
- Fat tails
- Skewness, kurtosis

Hands-on

- Fit Normal distribution
- Compare empirical vs Normal

Deliverable

- Notebook: *"Why Normal fails for returns"*

Week 4: Volatility Modeling

Concepts

- Volatility clustering
- Rolling vs EWMA

Hands-on

- Rolling volatility
- EWMA volatility (RiskMetrics)

Deliverable

- Notebook: *"Volatility clustering demo"*

Week 5: Correlation & Risk Concentration

Concepts

- Correlation breakdowns in stress

- Concentration risk

Hands-on

- Rolling correlation
- Stress correlation assumptions

Deliverable

- Notebook: *"Correlation in normal vs stress"*
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Week 6: Value at Risk (VaR) - Historical

Concepts

- VaR definition
- Confidence levels
- Holding period

Hands-on

- Historical VaR (single asset → portfolio)

Deliverable

- Notebook: *"Historical VaR engine"*
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Week 7: Parametric & Monte Carlo VaR

Concepts

- Parametric assumptions
- Simulation logic

Hands-on

- Parametric VaR
- Monte Carlo VaR

Deliverable

- Notebook: *"MC VaR vs Historical VaR"*
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Week 8: Backtesting

Concepts

- Exceptions
- Model accuracy vs conservatism

Hands-on

- VaR breaches
- Kupiec test

Deliverable

- Notebook: *"VaR backtesting results"*
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Week 9: Expected Shortfall (ES)

Concepts

- Why VaR fails
- Tail risk

Hands-on

- Historical ES
- MC-based ES

Deliverable

- Notebook: *"ES vs VaR comparison"*
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Week 10: Stress Testing

Concepts

- Historical stress
- Hypothetical stress
- Sensitivity vs scenario

Hands-on

- 2008-style shock
- Custom rate / FX shock

Deliverable

- Notebook: *"Stress P&L framework"*

 **You are now Market Risk-ready.**

Phase 2 - Model Risk (Market / Stress) (Weeks 11-14)

Week 11: Model Assumptions

Concepts

- Stationarity
- Window sensitivity
- Distribution choice

Hands-on

- Change VaR window length
- Compare outcomes

Deliverable

- Notebook: *"Model sensitivity analysis"*
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Week 12: Challenger Models

Concepts

- Benchmarking
- Model conservatism

Hands-on

- Compare Historical vs MC VaR
- Identify weaknesses

Deliverable

- Notebook: *"Challenger model comparison"*
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Week 13: Stress Model Validation

Concepts

- Severity vs plausibility
- Reverse stress testing

Hands-on

- Design implausible stress
- Explain why it's rejected

Deliverable

- Notebook: *"Stress model critique"*
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Week 14: Model Documentation

Concepts

- Model lifecycle
- Governance
- Limitations

Hands-on

- Write a **Model Validation Report**

Deliverable

- PDF/Markdown: *"Market Risk Model Validation Report"*

 **You are now MRM-capable.**

Phase 3 - Capstone Project (Weeks 15-16)

Integrated Project

 **Build a Market Risk Engine and Independently Validate It**